Math 54 — Week 9

Solutions to the problem set

P1. **Idea.** All four quantities come from dot products; the Pythagorean theorem is checked on $\|\mathbf{u}+\mathbf{v}\|^2$.

1. $\mathbf{u}\cdot\mathbf{v} = 2-2-3+3 = 0$, so $\mathbf{u}$ and $\mathbf{v}$ are orthogonal.
2. $\|\mathbf{u}\| = \sqrt{1+4+1+9} = \sqrt{15}$ and $\|\mathbf{v}\| = \sqrt{4+1+9+1} = \sqrt{15}$.
3. $\mathbf{u}+\mathbf{v} = (3,1,2,4)$, so $\|\mathbf{u}+\mathbf{v}\| = \sqrt{9+1+4+16} = \sqrt{30}$.
4. The Pythagorean theorem holds:

$$\|\mathbf{u}+\mathbf{v}\|^2 = 30 = 15+15 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2.$$

P2. **Idea.** $W^\perp$ is the null space of the matrix whose rows span W.

1. $\mathbf{x}\in W^\perp$ means $x_1+x_3+x_4=0$ and $x_2-x_3+2x_4=0$; this system is already in reduced echelon form with free variables x_3, x_4.
2. $x_1 = -x_3-x_4$ and $x_2 = x_3-2x_4$, so $\mathbf{x} = x_3(-1,1,1,0) + x_4(-1,-2,0,1)$.
3. Basis of $W^\perp$: $(-1,1,1,0)$, $(-1,-2,0,1)$. Check: $\dim W + \dim W^\perp = 2+2 = 4$, and each of the two basis vectors has dot product 0 with $(1,0,1,1)$ and with $(0,1,-1,2)$.

P3. **Idea.** Check the three dot products, then use $c_j = (\mathbf{x}\cdot\mathbf{b}_j)/(\mathbf{b}_j\cdot\mathbf{b}_j)$.

1. $\mathbf{b}_1\cdot\mathbf{b}_2 = 1-1+0 = 0$, $\mathbf{b}_1\cdot\mathbf{b}_3 = -1+1+0 = 0$ and $\mathbf{b}_2\cdot\mathbf{b}_3 = -1-1+2 = 0$. Three nonzero orthogonal vectors are independent (Thm. 4), so they are a basis of $\mathbb{R}^3$.
2. With $\mathbf{x}\cdot\mathbf{b}_1 = 4$, $\mathbf{x}\cdot\mathbf{b}_2 = 6$, $\mathbf{x}\cdot\mathbf{b}_3 = 6$ and $\mathbf{b}_1\cdot\mathbf{b}_1 = 2$, $\mathbf{b}_2\cdot\mathbf{b}_2 = 3$, $\mathbf{b}_3\cdot\mathbf{b}_3 = 6$:

$$c_1 = \frac{4}{2} = 2, \qquad c_2 = \frac{6}{3} = 2, \qquad c_3 = \frac{6}{6} = 1.$$

3. So $\mathbf{x} = 2\mathbf{b}_1 + 2\mathbf{b}_2 + \mathbf{b}_3$. Check: $(2,2,0)+(2,-2,2)+(-1,1,2) = (3,1,4)$.

P4. **Idea.** Project onto the line; the perpendicular part is the difference.

1. $\mathbf{y}\cdot\mathbf{u} = 3+2 = 5$ and $\mathbf{u}\cdot\mathbf{u} = 5$, so $\hat{\mathbf{y}} = \frac{5}{5}\mathbf{u} = (1,2)$.
2. $\mathbf{z} = \mathbf{y}-\hat{\mathbf{y}} = (2,-1)$, and $\mathbf{z}\cdot\mathbf{u} = 2-2 = 0$. So $\mathbf{y} = (1,2)+(2,-1)$.
3. The distance from $\mathbf{y}$ to the line is $\|\mathbf{z}\| = \sqrt{5}$.

P5. **Idea.** The two spanning vectors are orthogonal, so the projection formula applies directly.

1. $(1,1,0)\cdot(1,-1,1)=0$, so these vectors are an orthogonal basis of W.
2. $\mathbf{y}\cdot(1,1,0)=5$ and $(1,1,0)\cdot(1,1,0)=2$; $\mathbf{y}\cdot(1,-1,1)=3$ and $(1,-1,1)\cdot(1,-1,1)=3$. So
$$\operatorname{proj}_W \mathbf{y} = \frac{5}{2}(1,1,0)+1\cdot(1,-1,1) = \Big(\frac{7}{2},\ \frac{3}{2},\ 1\Big).$$
3. $\mathbf{z}=\mathbf{y}-\operatorname{proj}_W\mathbf{y}=(-\frac{3}{2},\frac{3}{2},3)$, orthogonal to both basis vectors ($-\frac{3}{2}+\frac{3}{2}=0$ and $-\frac{3}{2}-\frac{3}{2}+3=0$).
4. The distance from $\mathbf{y}$ to W is
$$\|\mathbf{z}\| = \sqrt{\frac{9}{4}+\frac{9}{4}+9} = \sqrt{\frac{27}{2}} = \frac{3\sqrt{6}}{2}.$$
The closest point of W is $\operatorname{proj}_W\mathbf{y}=(\frac{7}{2},\frac{3}{2},1)$ (best approximation theorem).

P6. **Idea.** $\mathbf{x}^\mathsf{T}A^\mathsf{T}A\mathbf{x}$ is $\|A\mathbf{x}\|^2$.

1. If $A^\mathsf{T}A\mathbf{x}=\mathbf{0}$, multiply on the left by $\mathbf{x}^\mathsf{T}$:
$$0=\mathbf{x}^\mathsf{T}A^\mathsf{T}A\mathbf{x}=(A\mathbf{x})^\mathsf{T}(A\mathbf{x})=\|A\mathbf{x}\|^2.$$
Only the zero vector has length 0, so $A\mathbf{x}=\mathbf{0}$.
2. Conversely, if $A\mathbf{x}=\mathbf{0}$ then $A^\mathsf{T}A\mathbf{x}=A^\mathsf{T}\mathbf{0}=\mathbf{0}$.
3. So the two null spaces have the same elements: $\operatorname{Nul}A^\mathsf{T}A=\operatorname{Nul}A$.

P7. **Idea.** Both parts come from $U^\mathsf{T}U=I$: take determinants for the first, and test $(UV)^\mathsf{T}(UV)$ for the second.

1. $U^\mathsf{T}U=I$ gives $\det(U^\mathsf{T})\det U=1$ (W5). Since $\det U^\mathsf{T}=\det U$, $(\det U)^2=1$ and $\det U=\pm1$.
2. UV is $n\times n$, and
$$(UV)^\mathsf{T}(UV)=V^\mathsf{T}U^\mathsf{T}UV=V^\mathsf{T}IV=V^\mathsf{T}V=I.$$
So UV is a square matrix with orthonormal columns (Thms. 6 and 7), that is, an orthogonal matrix.

P8. **Idea.** Use $U^\mathsf{T}U=I$ inside P^2; for the null space, cancel U the same way.

1. The columns of U are orthonormal, so $U^\mathsf{T}U=I$ and
$$P^2=U(U^\mathsf{T}U)U^\mathsf{T}=UIU^\mathsf{T}=P,\qquad P^\mathsf{T}=(U^\mathsf{T})^\mathsf{T}U^\mathsf{T}=UU^\mathsf{T}=P.$$
2. If $P\mathbf{x}=\mathbf{0}$, then $U(U^\mathsf{T}\mathbf{x})=\mathbf{0}$; multiplying by U^T gives $U^\mathsf{T}\mathbf{x}=\mathbf{0}$. Conversely $U^\mathsf{T}\mathbf{x}=\mathbf{0}$ gives $P\mathbf{x}=\mathbf{0}$. So $\operatorname{Nul}P=\operatorname{Nul}U^\mathsf{T}$, the set of vectors orthogonal to $\mathbf{u}_1$ and $\mathbf{u}_2$, which is $W^\perp$.
3. These vectors satisfy $x_1+x_3=0$ and $x_2=0$, so $\operatorname{Nul}P=\operatorname{Span}\{(1,0,-1)\}$. (Check with the matrix UU^T in the handout: its rows give $\frac{1}{2}x_1+\frac{1}{2}x_3=0$ and $x_2=0$.)